ESE 3123 Concept Primer

Thermo-Fluid Devices · KUET · read this first: the subject from zero, before the past-paper solutions

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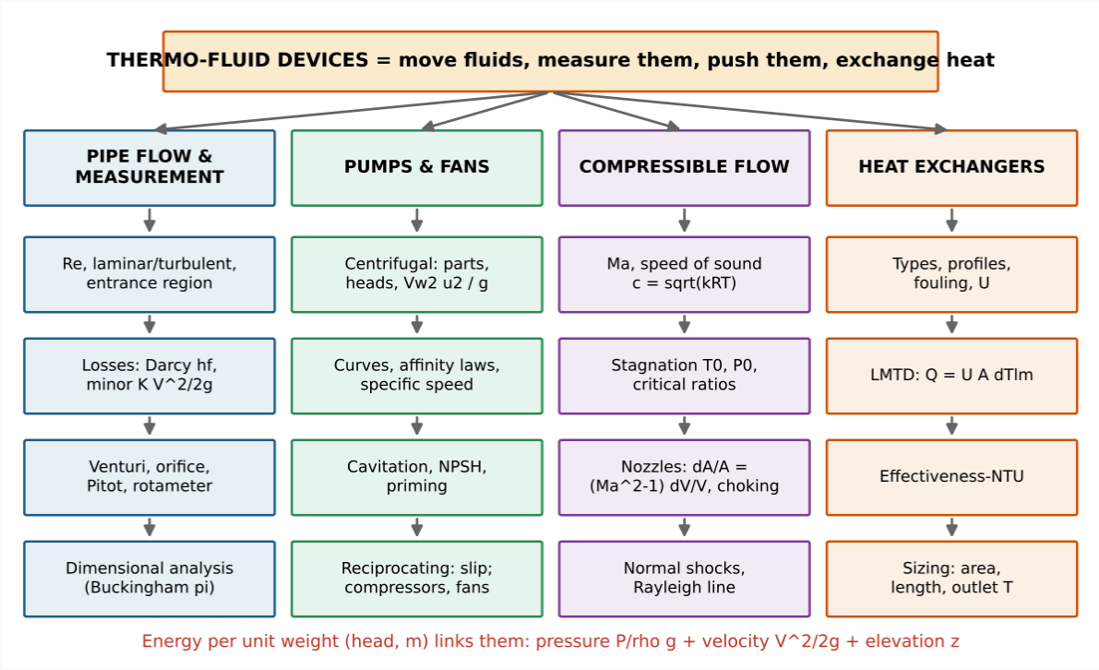
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How to use this primer

This booklet teaches thermo-fluid devices from zero, in the order the exam needs it. Read it once before the Complete Solutions. Each section has a green **core idea** box (learn this first), the formulas with every symbol and unit, a key figure and a short worked example. Section 2 is a maths and units refresher: most marks lost in this course come from unit slips (mm vs m, °C vs K, L/min vs m^3/s) and from using the wrong k or R for a gas.

The pass set (from the Question Map): pipe flow (A1, A2), flow meters (A4), centrifugal pumps (A5-A7), reciprocating pumps (A8), compressible flow and nozzles (B1, B2) and heat exchangers (B4-B6). The question numbers move between years (in 2019 and 2023 whole topics changed sections), so learn by topic. Dimensional analysis (A3), compressors and fans (A9) and shocks (B3) are the next to add.

The whole course on one page



Concept map of ESE 3123

Read the map in three blocks: **moving liquids** through pipes (losses) and **measuring** how much flows; **machines** that give the fluid energy (pumps, fans, compressors); and **hot and fast gases**: compressible flow in nozzles, then heat exchangers that pass heat from one fluid to another.

Maths and units refresher

> Convert everything to SI before you start: m, kg, s, N, Pa, W, K. Write the unit on every line. Keep 4 significant figures until the boxed answer.

Conversions you will meet

Quantity	Conversion
flow rate	1 L/s = 10^{-3} m^3/s; 1 L/min = 1.667×10^{-5} m^3/s; 1 gpm (US) = 6.309×10^{-5} m^3/s
pressure	1 bar = 10^5 Pa; 1 atm = 101.325 kPa; 1 m of water = 9.81 kPa; 1 mm Hg = 133.3 Pa
power	1 hp = 746 W (0.746 kW); 1 kW = 1000 J/s
length	1 in = 25.4 mm; 1 ft = 0.3048 m
temperature	T(K) = T(°C) + 273.15; a temperature **difference** is the same in K and °C
speed	rpm → rad/s: $\omega = 2\pi N/60$; blade speed $u = \pi DN/60$

Logarithms and exponentials

- ln is the natural log (base e); ln(a/b) = ln a − ln b; ln(1) = 0; if ln x = y then $x = e^y$.
- LMTD (log mean temperature difference): $(\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$. Either order gives the same answer (top and bottom both change sign).
- e^{-x} falls from 1 towards 0: $e^{-1} = 0.368$, $e^{-2} = 0.135$, $e^{-3} = 0.050$.

Powers and roots

- $x^a x^b = x^{a+b}$, $(x^a)^b = x^{ab}$, $x^{3/4} = (\sqrt{\sqrt{x}})^3$.
- Isentropic powers for air (k = 1.4): $k/(k-1) = 3.5$, $(k-1)/k = 0.2857$, $1/(k-1) = 2.5$, $(k-1)/2 = 0.2$.
- Gas constants: air R = 287 J/kg K, c_p = 1005; CO_2 R = 188.9, k = 1.289; helium R = 2077, k = 1.667.

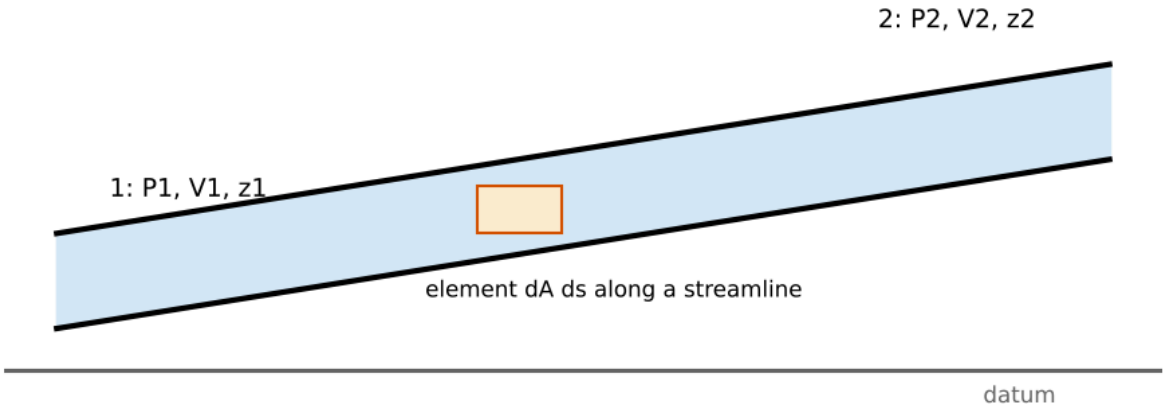
Dimensions

Every quantity can be written in M (mass), L (length), T (time): velocity LT^{-1}, force MLT^{-2}, pressure $ML^{-1}T^{-2}$, density ML^{-3}, viscosity μ $ML^{-1}T^{-1}$, power ML^2T^{-3}. A correct equation has the same dimensions in every term.

Fluid basics and Bernoulli's equation

> A flowing fluid carries three kinds of energy per unit weight ('heads', in metres): pressure head $P/\rho g$, velocity head $V^2/2g$ and elevation z. With no friction and no machine their sum is constant along a streamline (Bernoulli). Friction removes head (h_L); a pump adds head.

P/rho g + V^2/2g + z = constant (ideal, steady, incompressible, along a streamline)



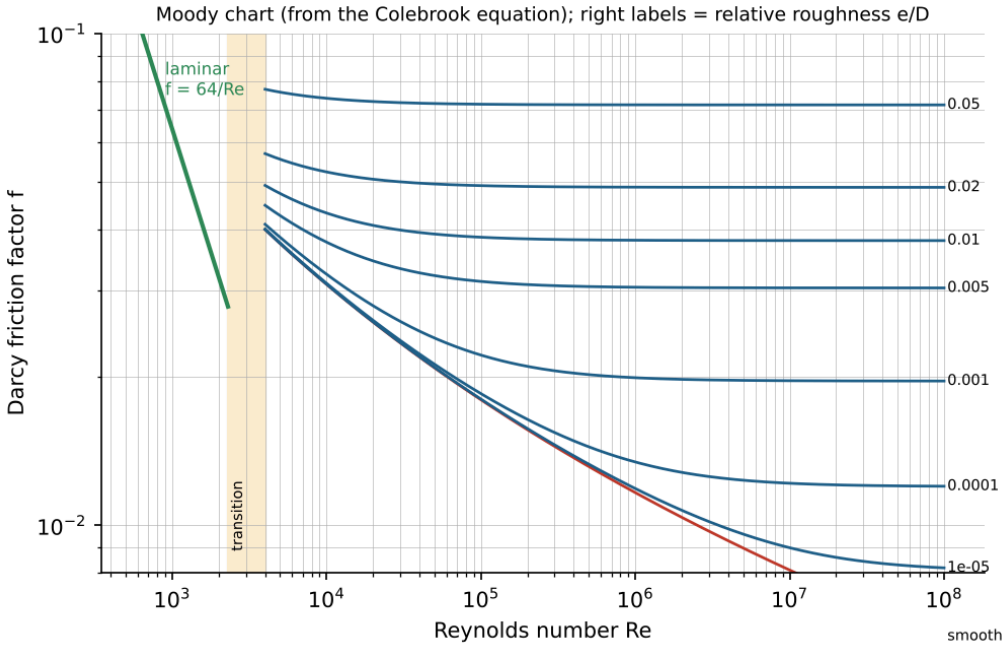
Bernoulli: the three heads along a pipe

$$P_1/\rho g + V_1^2/2g + z_1 + h_{pump} = P_2/\rho g + V_2^2/2g + z_2 + h_L; \quad Q = AV = \text{constant}$$

Symbol	Meaning	Unit
P	static pressure	Pa
ρ	density (water 1000)	kg/m^3
V	mean velocity	m/s
z	height above datum	m
h_L	head loss (friction)	m
Q	volume flow rate	m^3/s
μ, ν	dynamic viscosity, kinematic viscosity μ/ρ	Pa s, m^2/s

Flow in pipes (A1, A2)

The Reynolds number Re = ρVD/μ (inertia/viscous force) decides the type of flow: smooth layers (**laminar**, Re < 2300) or chaotic eddies (**turbulent**, Re > 4000). Friction head loss in a pipe is found by Darcy-Weisbach with a friction factor f: 64/Re if laminar, from the Moody chart if turbulent.



Moody chart (sketch)

$h_L = f\,(L/D)(V^2/2g)$; laminar $f = 64/Re$, $\Delta P = 32\mu LV/D^2$; minor loss $h_m = K\,V^2/2g$; pumping power $W = Q\Delta P$

Symbol	Meaning	Unit
f	Darcy friction factor	–
L, D	pipe length and inside diameter	m
ε/D	relative roughness	–
K	loss coefficient of a fitting	–
L_h	entrance (hydrodynamic) length	m

Example: water at 0.2 m/s in a 10 mm pipe, $\nu = 10^{-6}$: $Re = 0.2 \times 0.01/10^{-6} = 2000$ (laminar), $f = 0.032$, over 10 m: $h_L = 0.032 \times 1000 \times 0.04/19.62 = 0.065$ m.

Series and parallel pipes: in series the same Q passes and the losses add; in parallel each branch has the same loss and the flows add.

Dimensional analysis (A3)

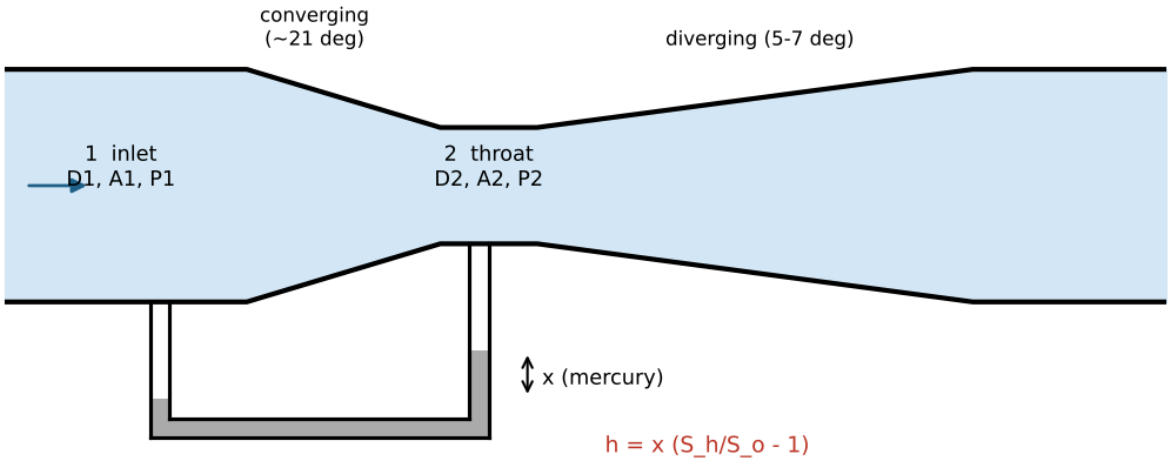
> If you know which variables a quantity depends on, you can group them into a few dimensionless numbers (π groups). This cuts the number of experiments and gives scale-model rules. Buckingham: n variables in m fundamental dimensions give n − m groups.

1. List the variables and their dimensions (M, L, T).
2. Choose m repeating variables that together contain M, L and T but do not form a group on their own (usually D, V, ρ).
3. Form each $\pi = D^a V^b \rho^c \times$(one other variable); solve the powers so π is dimensionless.
4. Write $\pi_1 = f(\pi_2, \pi_3, \ldots)$.

Example: drag F on a sphere depends on D, V, ρ, μ: n = 5, m = 3, 2 groups: $F/(\rho V^2 D^2) = f(\rho VD/\mu) = f(Re)$.

Measuring flow (A4)

> Most flow meters squeeze the flow (venturi, orifice, nozzle): velocity rises, pressure falls, and the measured pressure drop gives the velocity by Bernoulli. A Pitot tube instead stops the flow and reads the extra (stagnation) pressure. A rotameter and a turbine meter read flow directly from a float position or a rotor speed.



Venturimeter

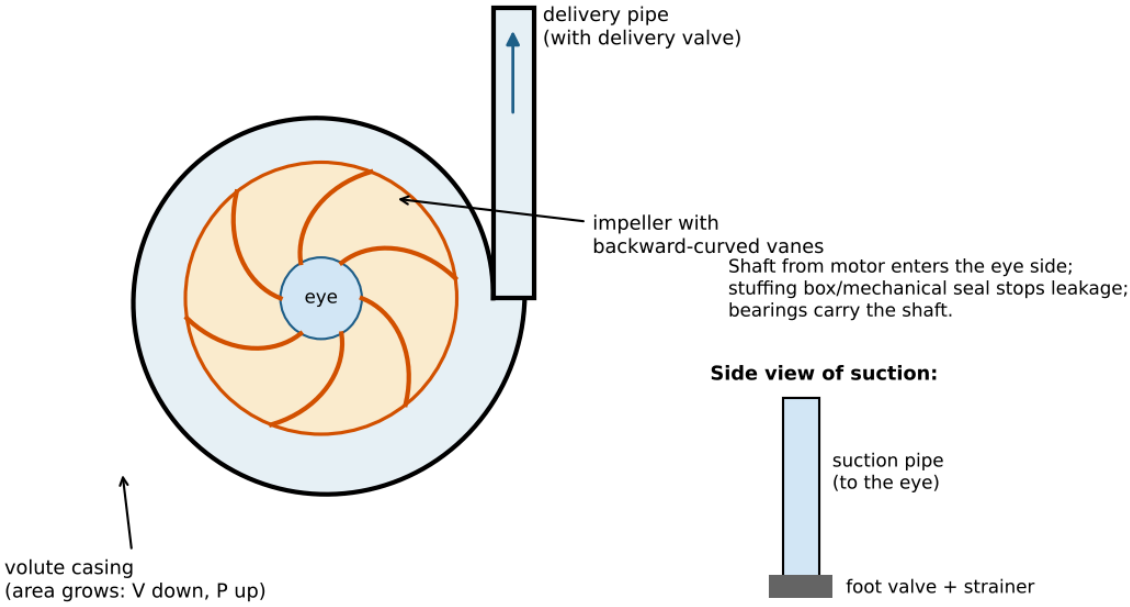
$Q = C_d\, a_1 a_2 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)}$; $h = x(S_m/S - 1)$; Pitot $V = C_v\sqrt{(2gh)}$

Symbol	Meaning	Unit
a_1, a_2	inlet and throat areas	m^2
C_d	coefficient of discharge (venturi 0.95-0.99, orifice about 0.62)	–
h	pressure difference as head of flowing fluid	m
x	manometer reading	m
S_m, S	specific gravity of manometer liquid and of flowing liquid	–

Example: mercury manometer reads x = 0.2 m on water: h = 0.2(13.6 − 1) = 2.52 m of water; a Pitot tube then gives V = √(2 × 9.81 × 2.52) = 7.03 m/s.

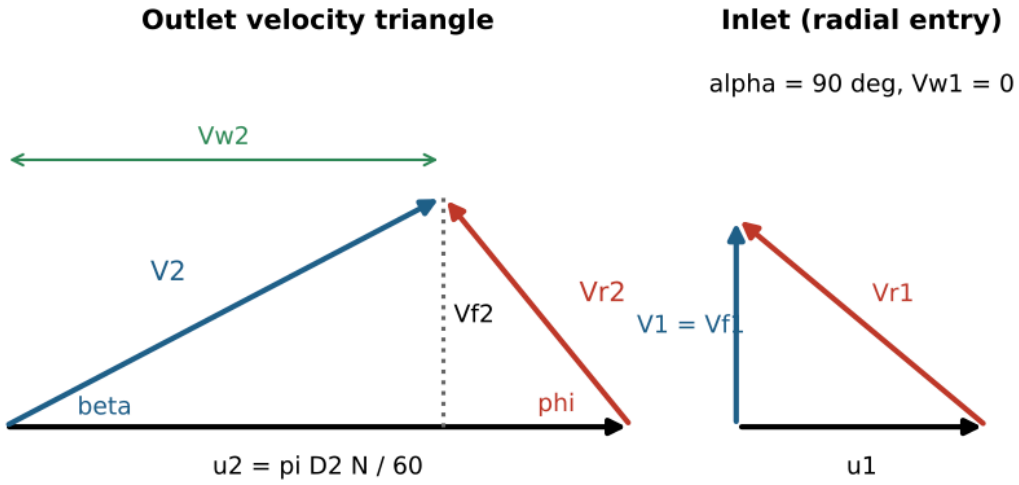
Centrifugal pumps (A5, A6, A7)

A rotating impeller throws the liquid outwards; the casing (volute or diffuser) turns its high speed into pressure. The head the pump gives depends on blade speed and outlet whirl velocity ($H = V_{w2}u_2/g$ ideally). Its performance at any speed follows the affinity laws, its type follows the specific speed, and it fails by cavitation if the suction pressure falls to the vapour pressure.



Centrifugal pump

Work per unit weight = (Vw2 u2 - Vw1 u1)/g = Vw2 u2 / g



Velocity triangles

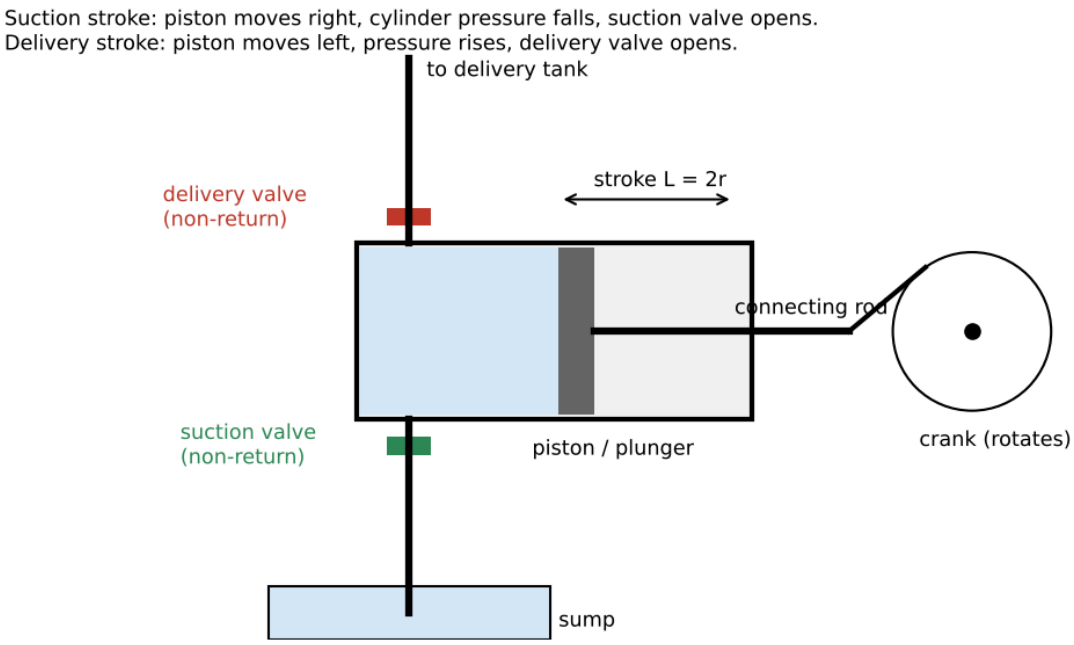
Work/weight = $V_{w2}u_2/g$; $n_{mano} = gH_m/(V_{w2}u_2)$; $P = \rho gQH/n_o$; $N_s = NvQ/H^{3/4}$; $Q \propto N$, $H \propto N^2$, $P \propto N^3$

Symbol	Meaning	Unit
u_2	blade tip speed $\pi D_2 N/60$	m/s
V_{w2}	whirl (tangential) component of outlet velocity	m/s
H_m	manometric head	m
N	speed	rpm
N_s	specific speed	rpm, m, m^3/s
NPSH (net positive suction head)	net positive suction head	m

Example: a pump gives 20 m at 1450 rpm; at 1800 rpm: $H = 20(1800/1450)^2 = 30.8$ m, and the flow rises by 1800/1450 = 1.24 times.

Reciprocating pumps, compressors and fans (A8, A9)

A reciprocating pump is a piston in a cylinder with suction and delivery valves: it moves a fixed volume per stroke, so it gives high pressure at low flow. The difference between the swept (theoretical) and actual delivery is the slip. Fans, blowers and compressors do for gases what pumps do for liquids; they are ranked by pressure ratio.



Reciprocating pump

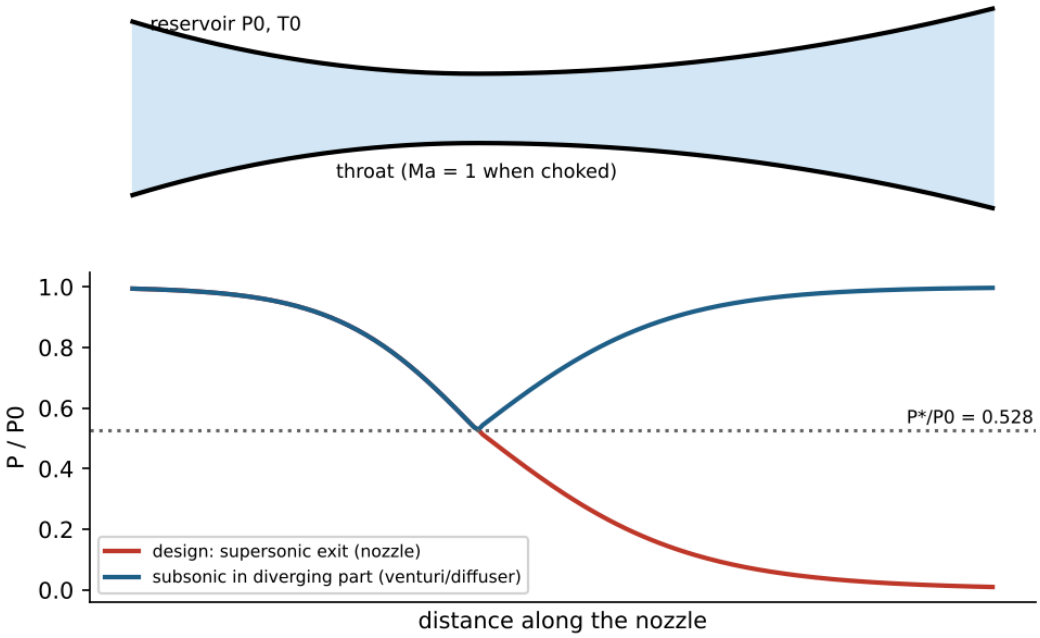
Q_{th} = ALN/60 (single acting), 2ALN/60 (double acting); $C_d = Q_{act}/Q_{th}$; slip = $Q_{th} - Q_{act}$; $P = \rho g Q_{th}(h_s + h_d)$

Symbol	Meaning	Unit
A	piston area $\pi D^2/4$	m^2
L	stroke length	m
N	crank speed	rpm
h_s, h_d	suction and delivery heads	m

Example: D = 0.15 m, L = 0.3 m, 60 rpm single acting: Q_{th} = 0.01767 × 0.3 × 1 = 5.30 L/s; if 5.0 L/s is delivered, C_d = 0.943 and slip = 5.7%.

Compressible flow and nozzles (B1, B2, B3)

When gas moves at a good fraction of the speed of sound, its density changes, and kinetic energy comes from its enthalpy: it cools as it speeds up. The Mach number Ma = V/c measures this. A subsonic flow speeds up in a converging duct, a supersonic flow in a diverging duct, so a rocket nozzle converges to a sonic throat and then diverges. Supersonic flow that must slow suddenly does it through a shock wave.



Converging-diverging nozzle

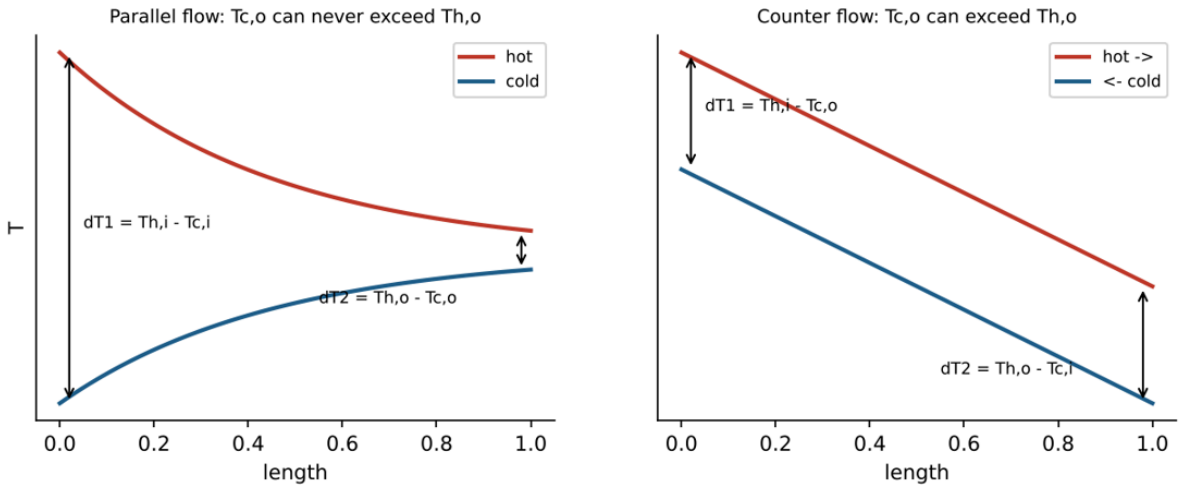
$c = \sqrt{kRT}$; $T_0/T = 1 + (k - 1)Ma^2/2$; $P_0/P = (T_0/T)^{k/(k-1)}$; $dA/A = (Ma^2 - 1)dV/V$; $\dot{m} = \rho AV$

Symbol	Meaning	Unit
Ma	Mach number V/c	–
c	speed of sound	m/s
k	specific heat ratio c_p/c_v (air 1.4)	–
R	gas constant (air 287)	J/kg K
T_0, P_0	stagnation temperature and pressure	K, Pa
$*$	critical (sonic, Ma = 1) state	–

Example: air from a tank at 500 K: at the throat T* = 0.8333 × 500 = 417 K and V* = √(1.4 × 287 × 417) = 409 m/s.

Heat exchangers (B4, B5, B6)

A heat exchanger passes heat from a hot fluid to a cold fluid through a wall. The rate is $Q = UA\Delta T_{lm}$, where U collects all the resistances and ΔT_{lm} is the proper average temperature difference. When outlet temperatures are unknown, use the effectiveness-NTU method instead. Counter flow always does better than parallel flow.



Temperatures along parallel-flow and counter-flow exchangers

$Q = \dot{m}_h c_{ph}(T_{h,i} - T_{h,o}) = \dot{m}_c c_{pc}(T_{c,o} - T_{c,i}) = UA\Delta T_{lm}$; $\varepsilon = Q/[C_{min}(T_{h,i} - T_{c,i})]$; NTU (number of transfer units) $= UA/C_{min}$

Symbol	Meaning	Unit
U	overall heat transfer coefficient	W/m^2K
A	heat transfer area πDL	m^2
ΔT_{lm}	log mean temperature difference	K or °C
C	heat capacity rate $\dot{m}c_p$	W/K
C_r	C_{min}/C_{max}	–
R_f	fouling factor	m^2K/W

Example: water 0.5 kg/s heated 20 → 60 °C by oil 120 → 80 °C in counter flow, U = 300: Q = 0.5 × 4180 × 40 = 83.6 kW; $\Delta T_1 = \Delta T_2 = 60$ °C so $\Delta T_{lm} = 60$; A = 83,600/(300 × 60) = 4.64 m^2.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
AODD	air-operated double diaphragm (pump)
ASME	American Society of Mechanical Engineers
BEP	best efficiency point
BHP	brake horsepower
C-D	converging-diverging
FD	forced draught
gpm	gallons per minute
hp	horsepower
HX	heat exchanger
ID	induced draught (or inside diameter)
kPa	kilopascal
LMTD	log mean temperature difference
MPa	megapascal
NPSH	net positive suction head
NTU	number of transfer units
rpm	revolutions per minute
TEMA	Tubular Exchanger Manufacturers Association
WHP	water horsepower
XXS	double extra strong (pipe schedule)